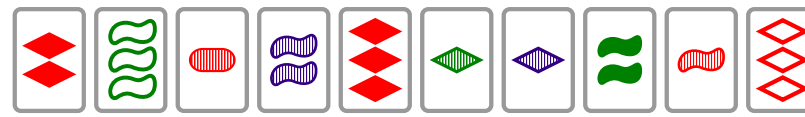
MathsJam Shout

May 2025
Duplicate SET Shout

What is SET?



SET is a game played with 81 cards that vary in four ways, with three possibilities for each: number (one/two/three), shape (diamond/squiggle/oval), shading (filled/hashed/empty), and colour (red/green/purple).

In the game, specific collections of three cards are said to make up a SET. For each of the four properties — number, colour, shading and shape — the three cards must either be **all the same**, or **all different**. Players compete to be the first to spot a SET from the cards on the table, and whoever spots the most wins.

This month, Guildford MathsJam have proposed a game a **Duplicate SET**, for any Jams who want to play along. It's locally non-competitive, and teams at each Jam will play together to try to get the best score. You may want to use a second deck for a practice run beforehand - but only have one go at the actual ordering.

Get SET

SET on Maths Gear: bit.ly/set-buy
A useful PDF: bit.ly/set-printable

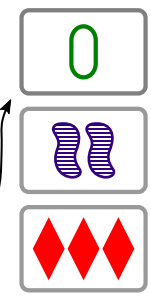
Rules Duplicate SET

Card order:
bit.ly/duplicate-SET

Set up a deck of cards in the given ordering (or do this before MathsJam to save time). Start a timer, then deal the first 12 cards and play the usual basic game, replacing any cards claimed as part of a SET by dealing three more from the top of the pile. If all players agree there's no SET, three additional cards can be dealt (returning to 12 cards once a SET(s) is found). Once you have matched 24 SETs, stop the timer and check your score.

Penalties and Bonuses:
Each MathsJam should field a team of 4 people. For each person over this on your team, add a **time penalty of 10 seconds**. If your team is fewer than 4 people, **deduct 10 seconds** per person from your time.

Sets that comprise 3 non-matched cards, i.e. the cards share no characteristic in common, are classed as '**Beasts**' or '**Level 4 SETs**'. These score a deduction of **5 seconds** from your overall time. Remember to keep these separate to make scoring easier.



To Discuss:
- What is the probability that it will be necessary to add three cards?
- What is the largest number of cards that can be revealed such that there is no set among them?

Final score: Total time in seconds for 24 sets + ((# people in team − 4) × 10) − (# Beast sets × 5)
Post your final result on Mastodon, or email to **katie@mathsjam.com**.

MathsJam Shout is a monthly sheet of ideas for activities to do at a MathsJam night. It's created using suggestions from a different MathsJam each month, and if you'd like to submit suggestions for a month in the future, email **katie@mathsjam.com** for details.

MathsJam is a monthly opportunity for like-minded self-confessed maths enthusiasts to get together in a pub and share stuff they like. Puzzles, games, problems, or just anything they think is cool or interesting. Monthly MathsJam nights happen in over 50 locations around the world, on the second-to-last Tuesday of each month. To find your nearest MathsJam, visit the website at **www.mathsjam.com**.